JESTER SURVIVABILITY

THREAT WARNING AND CMS ENHANCEMENT



A fan-made guide for a community mod. Not an official publication of Heatblur Simulations, Eagle Dynamics or the US Air Force. The emblem is original artwork.

START READ THIS FIRST

OVERVIEW

Jester Survivability is an add-on for the Heatblur F-4E's Jester AI. Stock Jester can sometimes fall short when it comes to situational awareness. This mod improves Jester's ability to see missiles based on their plume, and dispense countermeasures in a timely manner.

It also calls ground-launched radar missiles that the ALR-46 does not report.

This guide assumes you know the F-4E countermeasures panel and the ALR-46. It covers what the mod changes and what it cannot do.

YOU WILL NEED

- The DCS F-4E with Jester's mod loader (the `mods\init` folder).
- Nothing else for IR missiles and visible SAMs.
- Optional: the object feed (Section 1), for radar-guided missiles such as the AIM-7.

Written against DCS 2.9.30.

Thank you to Heatblur for making outstanding DCS modules, and for making the F-4 Jester API available to the community.

SYMBOLS

WARNING

Can cost you the aircraft.

CAUTION

Can break your install or open a security hole.

NOTE

Useful details.

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TERMS

Term	Meaning
Singer	An RWR or Jester call for a missile launch from a radar site.
Program	A countermeasure sequence: the number of releases, the interval between them, and how long each button press is held.
Time to impact	Seconds before a missile reaches the aircraft. The mod times every call and program against it.
Object feed	A small script that lists missiles fired by other coalitions, including the radar missiles DCS never gives Jester.
CMS	Countermeasures system: the chaff and flare dispenser and its panel.

SECTION 1 INSTALLATION AND THE OBJECT FEED

INSTALL

1. Close DCS. Open `Saved Games\DCS_F4E\jester\mods` (create the folders if they do not exist) and drag the folders `survivability` and `init` from the download into it. Choose **Copy and Replace** when Windows asks. The folders merge and only Survivability files are added.
2. Open `mods\survivability\Config.lua` and set `Config.SAVED_GAMES` (the line marked `PATH TO EDIT`) to your Saved Games folder, for example `A:\Saved Games`. `Config.SAVED_GAMES` is the only path to edit.
3. Start DCS. In a running mission, LShift+R restarts the mission and reloads Jester's Lua.

With only these files, Jester sees IR missiles and SAMs by their plumes, calls them, and dispenses. To remove the mod, delete `mods\init\survivability`; delete `mods\survivability` too for a full removal. The mod needs nothing from Jester A/A+ and can be installed alone or alongside it.

```
Saved Games\DCS_F4E\jester\mods\
    survivability\
        Suite, Countermeasures, ALE40, Log (.lua)
        Config.lua ← edit this one
        NOTES.md (developer notes)
    init\
        survivability\
            LaunchSurvivability.lua
```

FIG. 1-1. THE FINISHED INSTALL.

THE OBJECT FEED

DCS never gives Jester a semi-active radar missile (AIM-7, R-27R and R-27ER) in Jester's contact list, at any range. The only place such a missile can be read is DCS's object export. The feed is a short script that writes that list to a file for Jester to read.

	Feed off	Feed on
AIM-7, R-27R/ER	No call, no program	Called and countered
IR missiles	Called and countered	Called and countered
SAM missile	Called once Jester sees the plume	Called from the feed, once the plume is in view
SA-6 and other unreported SAMs	Singer call late or not at all	Singer call at launch

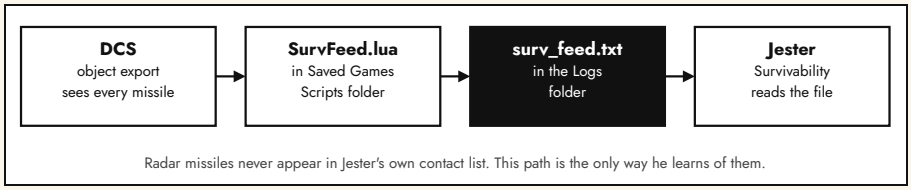


FIG. 1-2. HOW A RADAR MISSILE REACHES JESTER.

Set it up

1. Drag the folder `Scripts` from **_EXTRA (DCS Scripts)** into `Saved Games\DCS` (open beta: `Saved Games\DCS.openbeta`) and choose Copy and Replace. It adds `Scripts\SurvFeed.lua`.
2. Add this line at the end of `Saved Games\DCS\Scripts\Export.lua` (create the file if there is none):
   `pcall(function() local l=require("lfs"); dofile(l.writedir()..[[Scripts\SurvFeed.lua]]) end)`
3. In `<DCS install>\Scripts\MissionScripting.lua`, comment out the `sanitizeModule('io')` line. Jester needs `io` to read the file.
4. Restart DCS.

CAUTION

Commenting out `sanitizeModule("io")` gives every mission script access to the files on your PC. Do it only if you run missions you trust. A DCS update restores the line.

To check the feed, set `LOG_TO_FILE = true` in Config.lua. After a mission starts, `mods\survivability\survivability.log` holds a line reading `feed status: ... object export looks ON`. To remove the feed, delete the line from Export.lua and delete `SurvFeed.lua`.

SECTION 2 THREAT CALLS: MISSILES AND SINGERS

MISSILE CALLS

Jester sees a missile by its plume. The farther the missile, the longer Jester takes to pick out the plume. In a close fight Jester watches for a missile to come off the rail and calls it at once. A missile that has burned out and left no smoke is hard to see until it is close. Up to two calls follow.

1. **Early call, long shots only.** When Jester first sees an aircraft-fired missile more than 10 s from impact: "Missile, *clock*". SAMs get no early call.
2. **Warning.** At 14 s to impact (18 s for a SAM): "Missile missile missile, *clock*". Turn now; notch if the missile is radar-guided.

A shot launched inside those times gets the warning at once. The warning needs a missile that is closing and would pass within 1.5 NM of you. A long shot in the rear sector (4 to 8 o'clock) gets no warning and no program while it stays there: Jester assumes you have turned away. Countermeasures follow in Section 3.

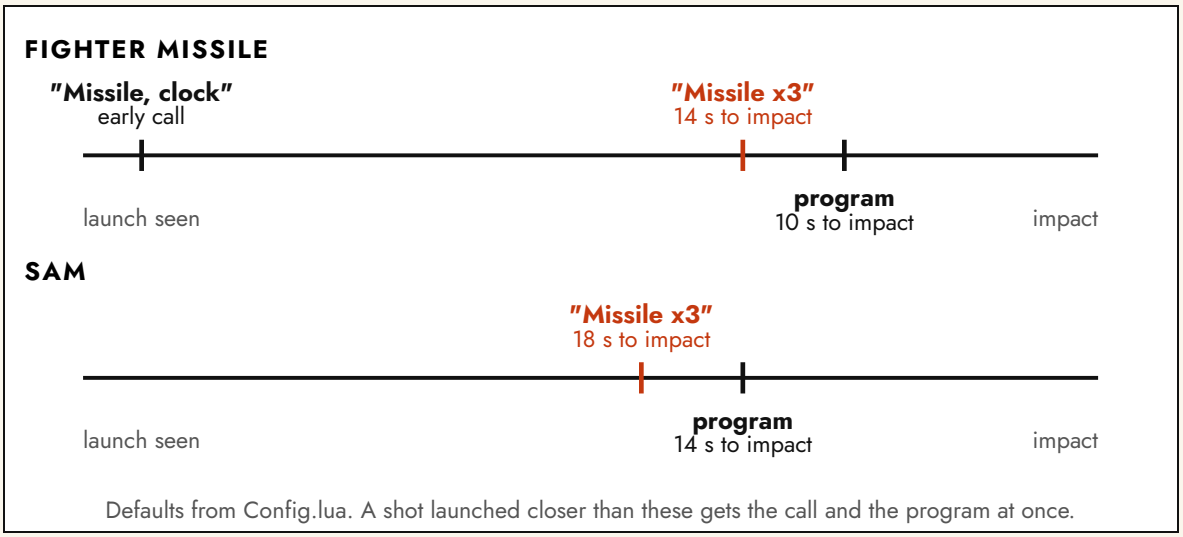


FIG. 2-1. WHEN THE CALLS AND THE PROGRAM FIRE. DEFAULTS SHOWN.

ONE CALL PER DIRECTION

Two missiles from one direction, or one missile reported twice (by the feed, then by Jester's eyes), give one call. A call within one clock position of the last, inside 8 s, is not repeated. A different direction is always called.

SINGER CALLS

The ALR-46 flags a launch only for a few systems, and stock Jester says nothing for the rest. For those, the mod makes its own call.

RWR reports a launch	No RWR launch indication
SA-2, SA-3, SA-15	SA-5, SA-6, SA-8, SA-10
NASAMS (tracking an AIM-120)	SA-11, SA-19
Active-radar air missiles: R-77, SD-10, AIM-54, AIM-120	Hawk, Patriot, Rapier

When Jester first sees a ground-launched radar missile, he calls "Singer, *clock*, *type*". The type comes from the RWR emitter at that clock position. If the RWR shows none, it comes from the missile's name. A call inside 5 s of another is spoken "and a singer". The same site is not called again for 30 s.

With the object feed, the call comes at launch. Without it, the call waits until Jester sees the missile; a missile Jester never sees gets no call.

Active-radar missiles. A missile that guides itself (AIM-54, AIM-120, R-77, SD-10) can be fired from beyond plume range, and a missile launched in TWS leaves no emitter on the RWR. Jester does not see it until the plume is in view, so a long-range Phoenix is called late. When the missile goes active the ALR-46 shows a missile symbol, and Jester answers it with "Missile missile missile, *clock*" and a chaff burst.

SECTION 3 COUNTERMEASURES AND SETUP ALE-40

WHICH PROGRAM

Jester dispenses at 10 s to impact (14 s for a SAM). A shot launched inside that time gets the program at once (Fig. 2-1 on the previous page). Countermeasures start for a closing missile that would pass within 3 NM of you.

Threat	Program
Close shot: within 7 NM, no RWR emitter at the shooter's bearing	Flares
Any other aircraft-fired missile	Chaff and flares
SAM	Chaff (chaff and flares if first seen close)
SAM launch shown only on the RWR, no missile seen	Chaff, 10 s after the launch

Program	Releases	Interval (s)	Hold (s)
Flare	5	0.1	0.07
Chaff	6	0.5	0.10
Both	5	0.3	0.10

Jester sets the chaff and flare mode knobs on the WSO's control unit (the rear cockpit) to SGL while the program runs and restores them afterwards.

SETUP ALE-40

The wheel gains a **Setup ALE-40** item under **Systems**. Choosing it sets the dispenser: chaff to 6 bursts at 0.4 s, flares and salvos to single, Dispense Program to Flares, WSO seat flare mode to OFF. Jester acknowledges with a Roger. Do this once before the mission starts.

1. Open the wheel, then **Systems**.
2. Choose **Setup ALE-40**.

NOTE

The mod runs its own timed dump and does not ask Jester for chaff, because a request to Jester arrives late.

JESTER CMS MENU

The stock **Countermeasures Dispensing** item on the wheel decides who dispenses. **Jester** (the default) lets the mod dispense with its own programs. **Manual** stops the mod from dispensing, and the dispense button is yours. Setup ALE-40 selects Jester.

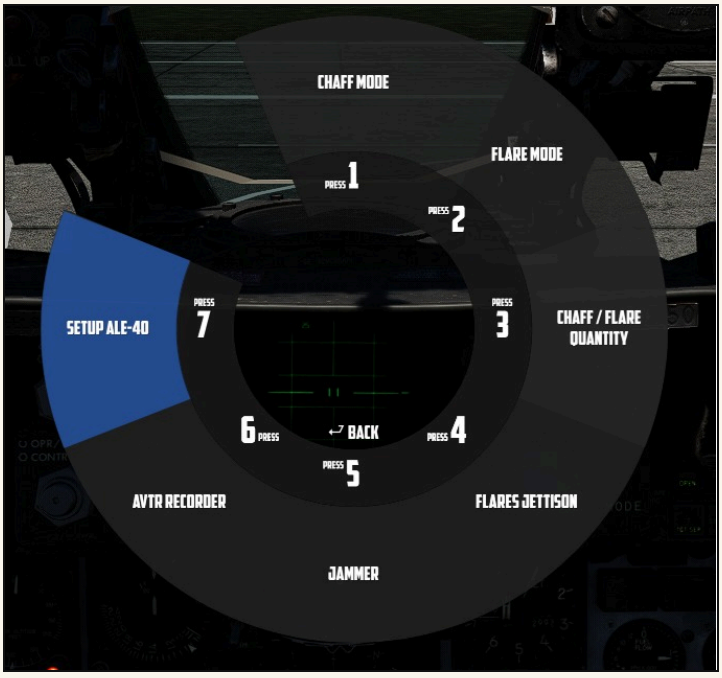


FIG. 3-1. THE WHEEL ITEM.

SECTION 4 LIMITS, TROUBLESHOOTING AND SETTINGS

LIMITS

- Radar missiles from aircraft get no call and no program without the object feed.
- The ALR-46 reports no launch for SA-5, SA-6, SA-8, SA-10, SA-11, SA-19, Hawk, Patriot or Rapier. The mod's singer call covers them.
- Jester speaks from fixed voice clips. The singer call always runs in the order singer, clock, type.
- The mod never asks Jester for chaff.
- A long-range active-radar missile such as the AIM-54 is called late unless the ALR-46 shows its missile symbol.

IF NOTHING HAPPENS

- Confirm `mods\init\survivability` exists.
- Set `LOG_TO_FILE = true` and read `mods\survivability\survivability.log`. The log needs the MissionScripting step in Section 1.
- For radar missiles, look for the feed status line.

LOG LINES

Line	Meaning
feed status: ... object export looks ON	The feed is live.
feed missile: / missile seen:	A missile was first tracked, with its program.
ground launch call (feed|eyes): ...	A singer call was made. "unknown" means no type was found: send the line to the author.
RWR launch: ...	The RWR showed a launch.

SETTINGS

All settings sit in `mods\survivability\Config.lua`, each with a comment. The common ones:

Setting	Default	Effect
SAVED_GAMES	your folder	Your Saved Games folder, for example A:\Saved Games. The only path to edit (Section 1).
AIR.warn_tti_s	14	Seconds to impact for the "missile missile missile" call.
SAM.warn_tti_s	18	Same, for SAMs.
AIR.dispense_tti_s	10	Seconds to impact for the program.
SAM.dispense_tti_s	14	Same, for SAMs.
PROGRAMS	Section 3	Releases, interval and hold for flare, chaff and both.
CALLS.gap_s	8	Window for the one-call-per-direction rule.
SINGER.same_site_s	30	No repeat singer call for one site inside this time.
LOG_TO_FILE	false	Writes survivability.log.

NOTE

COUNTERMEASURES and ALE40_SETUP turn the two halves of the mod on or off.

CARD QUICK REFERENCE

CALLS

Call	Means
"Missile, clock"	Long shot seen. Early warning.
"Missile missile missile, clock"	Turn now. Notch if the missile is radar-guided.
"Singer, clock, type"	Ground launch, from that site.

TIMING

	Warning	Program
Fighter missile	14 s	10 s
SAM	18 s	14 s

Print at 50% for a kneeboard card.

SETUP

Wheel, **Systems**, **Setup ALE-40**, before the mission starts.

FEED

Needed for AIM-7, R-27R/ER and similar semi-active radar missiles. Check the log for `object export looks ON`.

WARNING

Without the feed, Jester makes no call and dispenses nothing for radar missiles fired by aircraft.

NOTE

LShift+R restarts the mission and reloads Jester's Lua.